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Management (BDEEM)***

MASTER'S THESIS

**Market Basket Analysis and Temporal Variations of
Association Rules: An Analysis of Retail Transaction
Data**

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Digital Economics for Effective Management*

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General Introduction

Context

In recent years, the retail sector has undergone a major digital transformation. The expansion of supermarkets, e-commerce platforms, loyalty programs, and electronic payment systems has led to the continuous generation of large volumes of transactional information. Every purchase made by a customer leaves a digital trace that can provide valuable insights into consumer preferences and shopping habits.

The increasing availability of these data has created new opportunities for companies to improve their decision-making processes. Rather than simply storing information, organizations now seek to extract knowledge that can help them better understand their customers, optimize business operations, and gain a competitive advantage. This objective is closely related to the field of data mining, which focuses on discovering hidden patterns and relationships within large datasets.

Among the various data mining techniques, Market Basket Analysis (MBA) has become one of the most widely used approaches in the retail industry. This method examines customer transactions to identify products that are frequently purchased together. The relationships discovered through this analysis provide retailers with a better understanding of purchasing behavior and support more informed business decisions.

The practical applications of Market Basket Analysis are numerous. Retailers can use the identified product associations to improve store layout, design promotional bundles, optimize inventory management, and develop recommendation systems. For example, products that are often purchased together can be placed in nearby locations or included in the same marketing campaign to encourage additional purchases.

Although Market Basket Analysis provides valuable information about customer behaviour, it is often performed on the entire dataset without considering the temporal dimension. This global approach assumes that purchasing patterns remain relatively stable over

time. However, consumer behaviour is not always constant. Certain products may become more popular during specific seasons, holidays, or periods of the year. For instance, refreshing products are generally purchased more frequently during the summer months, while other categories may experience higher demand during colder seasons or festive periods.

As a result, analysing all transactions together may hide important temporal variations in customer purchasing behaviour. Some product associations may appear strongly during a limited period while remaining almost invisible in the overall analysis. Understanding these temporal dynamics can provide retailers with a more accurate view of consumer behaviour and contribute to more effective marketing and operational strategies.

Research Problem

Traditional Market Basket aims to identify relationships between products by analysing all customer transactions together. This global approach provides an overall view of purchasing behaviour and helps retailers discover common product combinations. However, it implicitly assumes that these relationships remain stable throughout the observation period.

In practice, customer purchasing behaviour is often influenced by temporal factors. Seasonal changes, holidays, weather conditions, and specific events can modify purchasing habits and create temporary associations between products. For example, products related to barbecues may be more frequently purchased together during the summer months, while hot beverages or comfort foods may become more strongly associated during colder periods. Similarly, major events and festive periods can generate specific purchasing patterns that are not observed during the rest of the year. For example, during international football competitions such as the FIFA World Cup, supporters may purchase national flags together with team jerseys and other fan merchandise, creating temporary product associations that disappear once the event is over.

Consequently, analysing all transactions together can hide these temporal variations. Product associations that are particularly strong during certain periods may appear weak when they are averaged over an entire year. Retailers who rely only on a global analysis may therefore

overlook important information that could improve promotional campaigns, inventory planning, product placement, and recommendation strategies.

This limitation raises an important research issue. If customer purchasing behaviour evolves over time, then association rules should not only be studied globally but also from a temporal perspective. Integrating the time dimension may reveal patterns that remain invisible in a traditional market basket analysis and provide a more accurate representation of customer behaviour.

Therefore, the central research question of this study is the following:

To what extent do association rules identified through Market Basket Analysis remain stable over time, and what additional managerial insights can be obtained by integrating a temporal dimension into the analysis?

To answer this question, the present study combines a traditional market basket analysis with a complementary temporal investigation to evaluate whether product associations evolve during different periods of the year and to assess the practical value of this additional information for retail decision-making.

Research Objectives

The main objective of this study is to analyse consumer purchasing behaviour through Market Basket Analysis using transactional retail data.

To achieve this objective, the study pursues four specific objectives:

- To prepare and transform transactional data into a format suitable for association rule mining.
- To identify frequent item sets and association rules that reveal relationships between products purchased together.
- To compare the use of Orange and Python for performing Market Basket Analysis on the selected dataset.

- To investigate whether association rules remain stable throughout the year or vary according to different periods, with particular attention to temporal variations in consumer purchasing behaviour.

These objectives provide the framework for the empirical analysis conducted in this study and guide the investigation of both global and temporal patterns in retail transactions.

Research Questions

Based on the research problem and objectives presented above, this study seeks to answer the following research questions:

RQ1: What product associations can be identified from retail transactional data through Market Basket Analysis?

RQ2: What are the respective strengths and limitations of Orange and Python for performing association rule mining?

RQ3: Do association rules identified through Market Basket Analysis remain stable throughout the year, or do they vary according to specific periods?

RQ4: What are the potential implications of temporal patterns in association rules for retail decision-making, particularly in relation to promotions, inventory management, and product placement?

Together, these research questions provide the foundation for investigating both the analytical and managerial dimensions of Market Basket Analysis in the retail sector.

Thesis Structure

This thesis is organized into six chapters.

General Introduction introduces the context of the study, presents the research problem, defines the research objectives, and formulates the research questions.

Chapter 1 reviews the academic literature related to the theoretical foundations of Market Basket Analysis and its applications in the retail sector. It also presents the key concepts and mathematical measures used throughout the study.

Chapter 2 describes the research methodology. It presents the dataset, the data preparation process, and the analytical approaches implemented using Orange and Python.

Chapter 3 presents the results of the empirical analysis, including the identification of product associations and the examination of temporal variations in association rules across different periods.

Chapter 4 discusses the findings in relation to the existing literature and highlights their managerial implications for the retail sector. It also addresses the limitations of the study and proposes directions for future research.

General Conclusion provides a synthesis of the study by answering the research questions, summarizing the main contributions, and presenting the overall conclusions.

Chapter 1. Literature Review

1.1 Data Mining and Knowledge Discovery

The retail sector generates large amounts of transactional data every day. Each customer purchase records information about the products bought, the date of the transaction, and other characteristics that can provide valuable insights into consumer behaviour. As retailers collect millions of transactions over time, the challenge is no longer the availability of data but the ability to transform these data into useful knowledge that can support decision-making.

This objective is closely related to the concept of Knowledge Discovery in Databases (KDD). According to Fayyad et al. (1996), KDD is the overall process of discovering valid, novel, useful, and understandable patterns in data. It involves several stages, including data selection, data cleaning, data transformation, data mining, and the interpretation of results. The purpose of this process is to convert raw data into knowledge that can support organizational objectives.

Within the KDD process, data mining represents the analytical stage in which algorithms are used to identify hidden patterns, relationships, and trends. Fayyad et al. (1996) describe data mining as a key step in the discovery of previously unknown and potentially useful patterns from large datasets. As a result, it has become an important tool for organizations that must analyse large and complex databases.

According to Raorane et al. (2012), data mining techniques allow organizations to extract knowledge from large transactional datasets and transform it into information that supports business decisions. In the retail sector, these techniques are particularly useful for understanding customer purchasing behaviour and identifying recurring patterns in transactions.

One important application of data mining in retail is the analysis of relationships between products purchased within the same transaction. Rather than focusing only on individual product frequencies, retailers are increasingly interested in understanding how products are associated with one another and how these associations reflect customer

purchasing behaviour. This objective led to the development of Market Basket Analysis, a technique specifically designed to identify associations between products.

A major contribution to this field was made by Agrawal et al. (1993), who introduced the concept of association rule mining for large transactional databases. Their work established the theoretical foundations for identifying relationships between items and inspired many subsequent developments in Market Basket Analysis. Because association rules constitute the core of Market Basket Analysis, it is necessary to examine this approach in greater detail.

1.2 Market Basket Analysis

Market Basket Analysis (MBA) is a data mining technique used to analyse customer transactions and identify products that are frequently purchased together. According to Raorane et al. (2012), the main objective of Market Basket Analysis is to extract useful knowledge from transactional data to better understand customer purchasing behaviour and support business decision-making. As a result, MBA has become one of the most widely applied analytical techniques in the retail sector.

The fundamental unit of analysis in Market Basket Analysis is the transaction, which corresponds to a customer's purchase containing one or more products. By analysing many transactions, it becomes possible to identify recurring purchasing patterns and relationships between products. Rather than focusing only on individual products, MBA seeks to understand how products are associated within the same shopping basket.

An important distinction must be made between product frequency and product association. Product frequency indicates how often a product appears in transactions, whereas product association examines whether products tend to appear together. A highly purchased product is not necessarily strongly associated with other products, while products with lower individual frequencies may exhibit strong relationships when purchased together.

The analysis of these relationships provides valuable information for retailers. According to Martinez et al. (2021), associations between products can support decisions related to promotional campaigns, product placement, and inventory management. By identifying products that are frequently purchased together, retailers can create more effective

promotional bundles and optimize merchandising strategies. Market Basket Analysis can also support recommendation systems and cross-selling initiatives. Prasetio et al. (2025) show that association rules can generate interpretable product recommendations by identifying meaningful relationships between products frequently purchased together. Unlike some complex recommendation models, association rules provide transparent and easily understandable results for decision-makers.

The theoretical foundations of Market Basket Analysis originate from the work of Agrawal et al. (1993), who introduced association rule mining for large transactional databases. Their framework made it possible to identify relationships between items through rules linking products that frequently appear together within transactions. These association rules remain the core analytical component of Market Basket Analysis and form the basis of most contemporary applications in retail analytics.

Because association rules are central to Market Basket Analysis, it is necessary to examine the mathematical measures used to evaluate them. The next section therefore introduces support, confidence, and lift, which are three most important measures for assessing the strength and relevance of product associations.

1.3 Mathematical Foundations

Association rule mining aims to identify meaningful relationships between products that frequently appear together in transactional data. However, not all association rules are equally relevant. Some rules may occur very frequently but provide little useful information, while others may represent strong relationships despite appearing less often. For this reason, several statistical measures are used to evaluate the quality and significance of association rules. According to Agrawal et al. (1993), support and confidence are the fundamental measures used in association rule mining. Hahsler et al. (2011) further emphasized the importance of lift as an additional measure for evaluating the strength of associations. Together, these indicators help identify the most relevant rules within a dataset.

1.3.1 Support

Support measures how frequently an association appears in the dataset. It represents the proportion of transactions containing both the antecedent and the consequent of a rule.

The support of a rule $(X \rightarrow Y)$ is calculated as:

$$\text{Support}(X \rightarrow Y) = \text{Transactions containing } XUY / \text{Total Transactions}$$

Where XUY represents transactions containing both the antecedent (X) and the consequent (Y).

A high support value indicates that the association occurs frequently within the dataset. Consequently, rules with very low support may be considered less relevant because they affect only a small number of transactions.

For example, if 50 transactions out of 1000 contain both products X and Y, the support of the rule is 0.05, meaning that the association appears in 5% of all transactions.

Support is particularly useful for identifying associations that are sufficiently frequent to be considered meaningful. However, support alone does not indicate whether the relationship between products is strong or reliable. Therefore, additional measures are required.

1.3.2 Confidence

Confidence evaluates the reliability of an association rule. It measures the probability that the consequent appears in a transaction when antecedent is already present.

The confidence of a rule $X \rightarrow Y$ is calculated as:

$$\text{Confidence}(X \rightarrow Y) = \text{Support}(XUY) / \text{Support}(X)$$

In other words, confidence corresponds to the conditional probability $(P(Y/X))$, which measures the likelihood of observing Y when X is already present in a transaction.

A confidence value of 0.50 indicates that 50% of the transactions containing X also contain Y.

For instance, a confidence value of 0.50 for the rule {sausage, whole milk} $\rightarrow$ {yogurt} indicates that 50% of the transactions containing sausage and whole milk also contain yogurt.

Confidence is useful because it indicates how often a rule is true. Nevertheless, a high confidence value does not necessarily imply a strong association. If the consequent is already very common in the dataset, confidence may be artificially inflated. Therefore, confidence should not be interpreted in isolation.

1.3.3 Lift

Lift is considered one of the most informative measures in association rule mining because it evaluates the strength of a relationship while considering the overall frequency of the consequent.

Lift is calculated as:

$$\text{Lift (X} \rightarrow \text{Y)} = \text{Confidence (X} \rightarrow \text{Y)} / \text{Support (Y)}$$

According to Hahsler et al. (2011), lift compares the observed occurrence of a rule with the occurrence that would be expected if the two products were statistically independent.

The interpretation of lift is straightforward:

- Lift = 1 indicates no association between the products.
- Lift > 1 indicates a positive association
- Lift < 1 indicates a negative association

When lift is greater than 1, the products are purchased together more frequently than expected by chance. Conversely, when lift is lower than 1, the products appear together less frequently than expected. Such situations may indicate substitution effects or products that are rarely purchased together.

For example, a lift value of 6 means that the consequent is six times more likely to occur when the antecedent is present than would be expected by chance alone.

In the context of retail analysis, lift is particularly valuable because it helps distinguish genuinely associations from those that merely reflect the popularity of individual products.

1.3.4 Complementarity of the Measures

Although support, confidence, and lift evaluate the same association rule, they provide different perspectives on its relevance and should therefore be interpreted together.

Support measures the frequency of an association, confidence measures its reliability, and lift measures its strength relative to random occurrence.

A rule may have high confidence but low lift, suggesting that the consequent is simply very common in the dataset. Similarly, a rule may have a high lift but extremely low support, limiting its practical usefulness. For this reason, association rule mining studies generally interpret these measures together when evaluating the relevance of discovered patterns.

The combined use of support, confidence, and lift provides a more comprehensive assessment of association rules and allows researchers to identify relationships that are both statistically meaningful and potentially useful for business decision-making.

1.4 Association Rule Mining Tools

Association rule mining relies on algorithms capable of identifying frequent item sets and generating association rules from large transactional datasets. As retail databases may contain thousands of transactions and hundreds of products, efficient algorithms are required to discover meaningful patterns within a reasonable computational time. Among the various algorithms proposed for association rule mining, Apriori (Agrawal and Srikant, 1994) and FP-Growth (Han et al. 2000) are widely recognized as two of the most influential and frequently applied approaches in both academic research and practical applications. Both aim to identify frequent item sets and generate association rules, but they differ significantly in their search strategy, computational efficiency, and memory requirements.

1.4.1 Apriori Algorithm

The Apriori algorithm, introduced by Agrawal and Srikant (1994), is one of the most influential algorithms in association rule mining. It was developed to efficiently identify frequent item sets in large transactional databases before generating association rules.

The algorithm is based on the Apriori principle, which states that if an itemset is frequent, all its subsets must also be frequent. Conversely, if an itemset is infrequent, all larger item sets containing it will also be infrequent.

For example, if the itemset {whole milk, yogurt} does not meet the minimum support threshold, then larger item sets such as {whole milk, yogurt, sausage} cannot be frequent and don't need to be evaluated.

This principle significantly reduces the search space and improves computational efficiency.

The Apriori algorithm operates through an iterative process:

1. Identify all frequent 1-itemsets.
2. Generate candidate 2-itemsets.
3. Retain only those meeting the minimum support threshold.
4. Generate larger candidate item sets from the frequent item sets identified in the previous iteration.
5. Repeat the process until no additional frequent item sets can be found.

After frequent item sets have been identified, association rules are generated and evaluated using measures such as support, confidence, and lift.

One of the main advantages of Apriori is its simplicity and interpretability. The algorithm is easy to understand and remains widely used in both academic research and practical applications. However, Apriori may become computationally expensive when applied to very large datasets because it requires multiple scans of the database and generates many candidate item sets.

1.4.2 FP – Growth Algorithm

To overcome some of the limitations of Apriori, Han and al. (2000) proposed the Frequent Pattern Growth (FP – Growth) algorithm.

Unlike Apriori, FP – Growth does not generate candidate item sets explicitly. Instead, it compresses the transactional database into a compact structure called an FP – Tree (Frequent Pattern Tree). This structure stores the frequency information of products while preserving the relationships between items. Transactions sharing common product prefixes are grouped together within the tree structure, which significantly reduces data redundancy and improves mining efficiency.

The algorithm proceeds in two main stages:

1. Construction of the FP Tree from the transactional dataset.
2. Extraction of frequent patterns directly from the tree structure.

By avoiding the generation of large numbers of candidate item sets, FP – Growth generally requires fewer database scans and can process large datasets more efficiently than Apriori.

As a result, FP – Growth is often preferred when working with large scale retail datasets containing many products and transactions. Nevertheless, the algorithm is more complex to understand and implement than Apriori, which may reduce its pedagogical value in some contexts.

1.4.3 Comparison of Apriori and FP – Growth

Although Apriori and FP – Growth pursue the same objective, they differ in several important aspects.

Criterion	Apriori	FP - Growth
Main principle	Candidate generation	FP – Tree construction
Database scans	Multiple scans	Fewer scans
Computational cost	Higher for large datasets	Lower for large datasets
Memory efficiency	Lower	Higher
Complexity	Simple to understand	More complex
Interpretability	High	Moderate

Apriori is generally considered easier to understand because its iterative process directly reflects the logic of association rule mining. FP – Growth, on the other hand, achieves greater computational efficiency by avoiding candidate generation and using a compressed tree structure.

Although FP – Growth is generally more efficient for large datasets, its tree-based structure is often less intuitive to interpret than the iterative candidate-generation process used by Apriori. Consequently, Apriori is frequently preferred in educational and explanatory contexts where transparency and ease of understanding are important.

The choice between the two algorithms often depends on the characteristics of the dataset and the objectives of the analysis.

1.4.4 Justification of the Algorithm Used in this Study

Both Apriori and FP – Growth played a role at different stages of this study.

During the exploratory phase, Orange Data Mining was used to identify frequent item sets and association rules. Orange relies on FP – Growth for frequent pattern extraction because of its computational efficiency and ability to handle large transactional datasets.

However, the main analysis was subsequently conducted in Python using the Apriori implementation available in the mlxtend library. This choice was motivated by several factors.

First, Apriori provides a transparent and easily interpretable framework for association rule mining. The step-by-step generation of frequent item sets facilitates understanding and

interpretation of the results. Second, Python offered greater flexibility in adjusting support and confidence thresholds, which proved particularly important for the analysis of a sparse retail dataset. Finally, Python enabled the implementation of the temporal analysis developed in this study, allowing association rules to be examined separately across different months.

Although FP – Growth is generally more computationally efficient, the size of the dataset used in this research remained manageable with Apriori. Consequently, the advantages of transparency, flexibility, and ease of interpretation outweighed the potential computational benefits of FP – Growth.

In summary, FP – Growth was useful during the exploratory stage conducted in Orange, whereas Apriori constituted the primary algorithm for the final analysis and interpretation of association rules in this study.

1.5 Retail Applications

Market Basket Analysis has become an important analytical tool in the retail industry because it enables organizations to transform transactional data into actionable business insights. By identifying products that are frequently purchased together, retailers can better understand purchasing behaviour and develop strategies aimed at increasing sales, improving operational efficiency, and enhancing customer satisfaction.

One of the most common applications of Market Basket Analysis is product placement. By identifying products that are frequently purchased together, retailers can optimize store layouts and place complementary items in nearby locations. This strategy increases the likelihood of additional purchases and improves the overall shopping experience. Association rule mining can support merchandising decisions by revealing relationships between products that may not be immediately visible through traditional sales analysis.

Market Basket Analysis is also widely used for cross-selling and product bundling. Cross-selling consists of recommending complementary products to customer based on their purchasing behaviour, while product bundling involves offering multiple products together as a package or promotional offer. By relying on association rules, retailers can identify combinations of products that are frequently purchased together and design more effective

promotional campaigns. Martinez et al (2021) highlight that such strategies can improve merchandising decisions and contribute to increased sales performance.

Another important application concerns inventory management. Understanding which products are commonly purchased together helps retailers anticipate demand more accurately and improve stock planning. When strong associations exist between products, inventory decisions can be coordinated to reduce stock shortages and maintain product availability. As a result, retailers can improve operational efficiency while ensuring that complementary products remain available when customers need them.

In recent years, Market Basket Analysis has also played an increasingly important role in recommendation systems, particularly in e-commerce environments. Association rules can be used to suggest products that are likely to interest customers based on items already selected or purchased. Prasetio et al (2025) emphasize that association-rule-based recommendation systems offer the advantage of transparency and interpretability, making recommendations easier to understand and justify than some complex machine learning approaches.

Beyond operational decisions, Market Basket Analysis provides valuable insights into customer behaviour and purchasing dynamics. By examining transaction patterns over time, retailers can identify recurring consumption habits, changing customer preferences, and seasonal purchasing behaviours. Rana and Mondal (2021) demonstrate that certain associations may only appear during specific periods of the year, suggesting that purchasing patterns are not necessarily constant over time. Similarly, Papavasileiou and Tsadiras (2011) show that the strength of association rules can vary significantly across different periods, while Kaur and Kang (2016) argue that consumer purchasing behaviour should be viewed as dynamic rather than static. These findings suggest that analyses conducted on aggregated datasets may conceal important temporal variations in customer behaviour.

The various retail applications of Market Basket Analysis illustrate its practical value beyond the identification of association rules. They also highlight the importance of considering how purchasing patterns evolve over time. This perspective is particularly relevant to the present study, which investigates whether association rules remain stable throughout the year and explores the potential implications of temporal variations for retail decision-making.

Chapter 2. Methodology

2.1 Dataset Description

The dataset used in this study was obtained from the Kaggle platform. The dataset consists of **38 765 observations**, each representing the purchase of a specific product by a customer on a given date. Although the dataset is publicly available, information regarding the original retailer, geographical location, and data collection process is not documented by the publisher. This limitation should therefore be acknowledged when interpreting the results.

The dataset contains the following variables:

- Member_number: unique identifier assigned to each customer
- Date: date on which the purchase was made
- ItemDescription: description of the purchased product
- Year: year of the transaction
- Month: month of the transaction
- Day: day of the month on which the transaction occurred
- day_of_week: day of the week associated with the transaction date

Overall, the dataset contains 169 distinct products, providing sufficient diversity for the identification of frequent item sets and association rules.

The date-related variables included in the dataset facilitate the analysis of purchasing behaviour over time and are particularly useful for investigating temporal variations in association rules. While the monthly analysis constitutes the primary temporal focus of this study, the additional date variables were retained as part of the original dataset structure and may support further temporal investigations in future research.

Furthermore, a new variable called Transaction_ID was created by combining the customer identifier and the transaction date. This transformation was necessary because a single shopping basket may contain multiple products purchased by the same customer on the same

day. The Transaction_ID therefore allows all products belonging to the same basket to be grouped into a single transaction for Market Basket Analysis.

After grouping purchases by customer and transaction date, the dataset contained **14 963 unique transactions**, each representing a complete shopping basket.

The dataset was selected primarily because it includes transaction dates, making it possible to investigate the temporal stability of association rules. This characteristic is particularly important for addressing the research questions of this study, which seek to determine whether purchasing patterns remain stable throughout the year or vary across different periods. In addition, the dataset contains a substantial number of transactions and a diverse range of grocery products, providing a suitable basis for association rule mining.

The data preparation procedures applied to the dataset are presented in the following section.

2.2 Data Preparation

Before applying association rule mining, the dataset had to be reorganized into a structure in which each transaction represented a complete shopping basket. This step was essential because Market Basket Analysis does not analyse isolated product lines, but combinations of products purchased within the same transaction.

As described in the previous section, a Transaction_ID variable was created by combining the customer identifier (Member_number) and the transaction date. This variable made it possible to group all products purchased by the same customer on the same day into a single basket. After this transformation, the dataset contained 14 963 unique transactions.

For the exploratory analysis in Orange Data Mining, the data was transformed into a basket matrix using a pivot table in Microsoft Excel. This choice was made because Orange requires a structured format in which each row corresponds to a transaction and each column corresponds to a product. In this matrix, the value 1 indicated that a product was present in a transaction, while the value 0 initially indicated that the product was absent.

During the first import into Orange, a formatting issue was identified. When product absence was coded as 0, Orange interpreted both 0 and 1 as valid categorical values. This was not appropriate for Market Basket Analysis, because the objective was to analyse product presence rather than product absence. Therefore, the absence values coded as 0 were replaced by the symbol “?”, which Orange treats as a missing value. This transformation allows Orange to focus on purchased products.

After this correction, the Frequent Item sets widget was able to extract frequent item sets. However, the Association Rules widget did not generate any rules. Despite testing multiple support and confidence thresholds, including the minimum confidence threshold available in Orange, no association rules were produced. This result suggests that the sparse structure of the dataset and the relatively weak associations between products limited the ability of Orange to identify valid rules under the available parameter settings. Consequently, Python was used for the main analysis, as it provided greater flexibility for parameter customization and rule extraction.

For the Python implementation, the original transactional dataset containing Transaction_ID was retained instead of the Excel pivot table. The preparation steps were performed directly in Python. First, the data was grouped by Transaction_ID and itemDescription to determine whether each product appeared in a transaction. The grouped data was then transformed into a transaction-product matrix, where rows represented transactions and columns represented products. Missing values were replaced with 0, and all positive values were converted to 1, producing a binary representation of product presence and absence. Finally, the matrix was converted into a Boolean format required by the association rule mining functions used in Python.

This Python-based preparation made the analysis more flexible and reproducible. It allowed customized support and confidence thresholds to be applied and made it possible to conduct both the global analysis and the monthly temporal analysis using the same preparation logic.

The exploratory analysis conducted in Orange Data Mining is presented in the following section.

2.3 Exploratory Analysis with Orange

Orange Data Mining is an open-source visual data mining software that allows users to perform exploratory analyses through graphical workflows. The software has been widely used for educational and data mining purposes because of its accessibility and interactive visual environment (Demšar et al. 2013). Orange Data Mining was used as an exploratory tool to perform an initial investigation of the prepared dataset before conducting the main analysis in python. The objective of this exploratory phase was to verify the structure of the data, identify frequent item sets, and evaluate the ability of Orange to generate association rules from the transactional dataset.

The prepared dataset was first imported into Orange using the File widget. To ensure that the data had been correctly formatted, the dataset was examined through the Data Table widget. This verification confirmed that product presence was represented by the value 1, while product absence was represented by the symbol “?”.

Before running the analysis, the product variables were manually converted from numeric to categorical attributes in Orange. During the initial tests, the Frequent Itemsets widget was unable to generate results because many product variables were interpreted as numeric rather than categorical. Since association rule mining algorithms operate on categorical items representing product presence or absence, the variables were redefined as categorical attributes. After this modification, the Frequent Itemsets widget was able to successfully generate frequent item sets.

The workflow implemented in Orange consisted of the following sequence: **File -> Data Table -> Frequent Itemsets -> Association Rules**. A screenshot of this workflow is provided in Appendix A.

After verifying the dataset structure, the Frequent Itemsets widget was applied. A minimum support threshold of 0.001 was selected to capture many potentially relevant product combinations. The analysis successfully generated frequent item sets and reached the maximum output limit of 100 000 item sets imposed by the software. This result confirmed that the dataset contained numerous recurring product combinations and was suitable for Market Basket Analysis.

The frequent item sets also confirmed the presence of recurrent purchasing patterns in the dataset. Whole milk appeared in 14.91% of the transactions and was observed in baskets of different sizes, ranging from two to four products. Yogurt was less frequent, appearing in approximately 1.02% of the transactions and occurring in baskets containing between two and eight products. These results indicate that some products were repeatedly purchased and distributed across baskets of different sizes.

The extracted item sets were then connected to the Association Rules widget. To maximize the possibility of obtaining association rules, the minimum confidence threshold was set to 1%, which corresponds to the lowest value allowed by Orange. Despite testing different support and confidence settings, including the minimum confidence threshold available, no association rules were generated. Although Orange did not generate association rules, the frequent item sets confirmed the suitability of the dataset for Market Basket Analysis.

The most likely explanation for this result is the sparse structure of the dataset, where most products appear in only a small proportion of transactions. This characteristic reduces the frequency of many product combinations and makes it more difficult to satisfy confidence requirements. In addition, the minimum confidence threshold available in Orange lay still have been too restrictive for the characteristics of the dataset.

Although Orange successfully identified frequent item sets, it did not generate association rules. Nevertheless, this exploratory phase was valuable because it confirmed the suitability of the dataset for Market Basket Analysis, highlighted the limitations of the software in this context, and guided the methodological decision to conduct the main analysis in Python, where greater flexibility in parameter selection was available.

2.4 Python Implementation

Following the exploratory analysis conducted in Orange Data Mining, Python was used as the main analytical environment for extracting association rules. Python was selected because it provides greater flexibility in parameter selection and allows more advanced analyses than those available in Orange. It enabled the extraction of association rules using customized thresholds and facilitated the implementation of a temporal analysis of purchasing patterns.

The analysis was performed using the pandas library for data manipulation and the mlxtend library for association rule mining. The Apriori algorithm was applied through the `apriori()` function, while association rules were generated using the `association_rules()` function.

As described in Section 2.2, the transactional data was transformed into a Boolean transaction-product matrix suitable for association rule mining. This matrix served as the input for all analyses performed in Python.

The Apriori algorithm was applied to identify frequent item sets. A minimum support threshold of 0.001 was selected to capture relatively infrequent but potentially meaningful product combinations. The resulting frequent item sets served as the basis for association rule generation.

Association rules were extracted using the confidence metric with a minimum confidence threshold of 0.001. The same minimum threshold was used for both support and confidence. Given the sparse structure of the dataset and the large number of distinct products, higher thresholds would have eliminated many potentially relevant associations. The selected values therefore allowed less frequent but potentially meaningful purchasing patterns to be identified. For each rule, several indicators were calculated, including support, confidence, and lift. These measures made it possible to evaluate the frequency, reliability, and strength of the relationships between products.

In addition to the global analysis, a temporal analysis was conducted at the monthly level. The dataset was divided into twelve subsets corresponding to the twelve months of the year. For each month, transactions were first filtered according to the corresponding month and then transformed into a transaction-product matrix. The matrix was converted into a Boolean format before applying the Apriori algorithm and generating association rules. The same thresholds (minimum support = 0.001 and minimum confidence = 0.001) were maintained throughout the analysis to ensure comparability across months.

To investigate the stability of purchasing patterns over time, the rule {whole milk, yogurt} $\rightarrow$ {sausage} was selected for monthly monitoring. This rule was chosen because it was identified in the global analysis, involved products that appeared frequently in the dataset, and was present in several monthly subsets. These characteristics made it appropriate for examining how support, confidence, and lift varied throughout the year. For each monthly dataset, these three indicators were extracted and compared to assess the temporal stability of the relationship.

These procedures enabled the identification of frequent item sets, association rules, and temporal variations in purchasing behaviour. The results of both the global and monthly analyses are presented in the following chapter.

Chapter 3. Results

3.1 Frequent Item sets

The first stage of the analysis consisted of identifying frequent item sets using the Apriori algorithm. In Market Basket Analysis, an itemset refers to a set of one or more products that appear in transactions. Table 3.1 presents the item sets with the highest support values in the dataset. Interestingly, the most frequent item sets were all single-product item sets, indicating that individual products appeared more frequently than any specific product combination.

The results indicate that the most frequently purchased products were whole milk (support = 15.79%), other vegetables (12.21%), rolls/buns (11.00%), soda (9.71%), and yogurt (8.59%). Other frequently purchased products included root vegetables (6.96%), tropical fruit (6.78%), bottled water (6.07%), sausage (6.03%), and citrus fruit (5.31%).

These results show that the most common purchases were everyday grocery products that are frequently consumed and regularly replenished by customers. Dairy products, vegetables, bakery products, beverages, and fruits were among the most recurrent items in shopping baskets, indicating their importance in customer's purchasing behaviour.

The prominence of whole milk is particularly noteworthy, as it appeared in 15.79% of all transactions, making it the most frequent item in the dataset. The frequent occurrence of whole milk, yogurt, and sausage is especially relevant for the subsequent analyses, as these products are involved in some of the association rules identified later in the study, including the rule selected for temporal monitoring.

Overall, the frequent itemset analysis confirms the existence of recurring purchasing patterns within the dataset and provides a solid foundation for the extraction and interpretation of association rules. The next section examines these relationships through the analysis of global association rules.

	support	itemsets
146	0.157923	(whole milk)
90	0.122101	(other vegetables)
109	0.110005	(rolls/buns)
123	0.097106	(soda)
147	0.085879	(yogurt)
110	0.069572	(root vegetables)
139	0.067767	(tropical fruit)
10	0.060683	(bottled water)
115	0.060349	(sausage)

Table 3.1. Item sets with the Highest Support values

3.2 Global Association Rules

After identifying frequent item sets, association rules were generated using the confidence metric and ranked according to their lift values. Lift was selected as the primary indicator for interpretation because it measures the strength of the relationship between products beyond what would be expected by chance.

Table 3.2 presents the association rules with the highest lift values obtained from the global analysis. The results reveal several positive associations between products, indicating that certain combinations of items were purchased together more frequently than expected under independence.

Among the strongest associations identified was the rule {whole milk, yogurt} -> {sausage}, which achieved a support of 0.147%, a confidence of 13.17%, and a lift of 2.18. This lift value indicates that customers who purchased both whole milk and yogurt were more than twice as likely to purchase sausage compared with what would be expected if these products were purchased independently. However, the support of this rule remained relatively low (0.147%), indicating that although the association was strong, it occurred in a limited number of transactions.

A similar pattern can be observed in the rule {sausage, whole milk} -> {yogurt}, which achieved a lift value of 1.91. Likewise, the rule {sausage, yogurt} -> {whole milk} obtained a lift of 1.62. These findings suggest a recurring relationship among whole milk, yogurt, and sausage, as the same products appear repeatedly in several of strongest association rules.

Other notable associations included the rule {specialty chocolate} -> {citrus fruit}, which achieved a lift value of 1.65, and its reverse rule {citrus fruit} -> {specialty chocolate}. Although the support of these rules was relatively low, their lift values indicate that these products were purchased together more frequently than expected under independence.

Overall, the results reveal statistically strong associations between specific products. Nevertheless, the relatively low support values indicate that these relationships occurred in a limited proportion of transactions and should therefore be interpreted with caution. The repeated presence of whole milk, yogurt, and sausage among the strongest rules highlights their importance within customer purchasing patterns. Because this combination appeared consistently among the strongest associations identified in the global analysis, the rule {whole milk, yogurt} -> {sausage} was selected for the temporal analysis presented in the next section.

	antecedents	consequents	support	confidence	lift
1236	(sausage)	(whole milk, yogurt)	0.001470	0.024363	2.182917
1233	(whole milk, yogurt)	(sausage)	0.001470	0.131737	2.182917
1232	(whole milk, sausage)	(yogurt)	0.001470	0.164179	1.911760
1237	(yogurt)	(whole milk, sausage)	0.001470	0.017121	1.911760
474	(citrus fruit)	(specialty chocolate)	0.001403	0.026415	1.653762
475	(specialty chocolate)	(citrus fruit)	0.001403	0.087866	1.653762

Table 3.2. Association Rules with the Highest Lift Values

3.3 Monthly Transaction Analysis

To investigate whether purchasing patterns remained stable throughout the year, the association rule {whole milk, yogurt} $\rightarrow$ {sausage} was analysed monthly. For each month, support, confidence, and lift were calculated using the same thresholds applied in the global analysis.

Table 3.3 presents the monthly results obtained for the selected rule. The findings reveal substantial variations across months, indicating that the relationship between whole milk, yogurt, and sausage was not constant throughout the year.

The rule was identified in only five months: January, March, April, September, and November. In the remaining months, no association rule matching the selected combination was generated. This result suggests that the co-purchasing behaviour linking these products was not consistently observed across all periods of the year.

The highest support value was recorded in January (0.465%), followed by November (0.393%). In contrast, support values remained relatively low in March, April, and September. These results indicate that the selected product combination occurred more frequently in January and November than in the other months where the rule was present. Nevertheless, all support values remained relatively low, indicating that the rule was observed in only a limited proportion of monthly transactions.

Confidence values also varied considerably. The highest confidence was observed in November (41.67%), meaning that when whole milk and yogurt were purchased together during this month, sausage was also purchased in approximately four out of ten cases. January displayed the second highest confidence value (31.58%), whereas March, April, and September showed substantially lower values.

Similar variations can be observed for lift. The strongest association was found in November, with a lift value of 8.69, followed by January with a lift of 5.36. In contrast, the lift values recorded in March (1.37), April (2.35), and September (2.10) were considerably lower. Although all observed lift values remained above 1, indicating positive associations, the strength of the relationship differed substantially across months.

Overall, the monthly analysis demonstrates that the association between whole milk, yogurt, and sausage was not stable throughout the year. The presence of the rule in only a limited

number of months, combined with the substantial variations in support, confidence, and lift, suggests that purchasing patterns may vary across different periods of the year. These monthly variations are examined further, in the following section.

	month	support	confidence	lift
0	1	0.004648	0.315789	5.364266
1	2	0.000000	0.000000	0.000000
2	3	0.001621	0.095238	1.366556
3	4	0.002451	0.136364	2.350832
4	5	0.000000	0.000000	0.000000
5	6	0.000000	0.000000	0.000000
6	7	0.000000	0.000000	0.000000
7	8	0.000000	0.000000	0.000000
8	9	0.001748	0.117647	2.102941
9	10	0.000000	0.000000	0.000000
10	11	0.003931	0.416667	8.688525
11	12	0.000000	0.000000	0.000000

Table 3.3. Monthly Performance of the Rule {whole milk, yogurt} -> {sausage}

3.4 Visualizing Temporal Variations

To further examine the monthly variations identified in the previous section, Figure 3.1 presents a bar chart showing the evolution of the lift value for the rule {whole milk, yogurt} -> {sausage} across the year.

The figure provides a clearer visual representation of the instability of the rule over time. The association was observed only during five months: January, March, April, September, and November. For the remaining months, the rule was not generated. Therefore, the absence of bars for these months should not be interpreted as a lift value equal to zero, but rather as the absence of the selected rule in the monthly association rule output.

The bar chart highlights a clear contrast between periods where the association was relatively weak and periods where it was substantially stronger. Visual inspection reveals pronounced peaks during January and November, whereas much lower levels were observed in March, April, and September.

This visual pattern suggests that the association between whole milk, yogurt, and sausage was not evenly distributed across the year. The stronger lift values observed during certain periods may indicate temporal variations in purchasing behaviour, although this interpretation should remain cautious because the support values relatively low. This point will be discussed further in Chapter 4.

Overall, Figure 3.1 reinforces the conclusion that the selected association rule was not stable over time. While the global analysis identified the rule as one of the strongest associations in the dataset, the monthly visualization shows that its strength varied considerably depending on the period analysed.

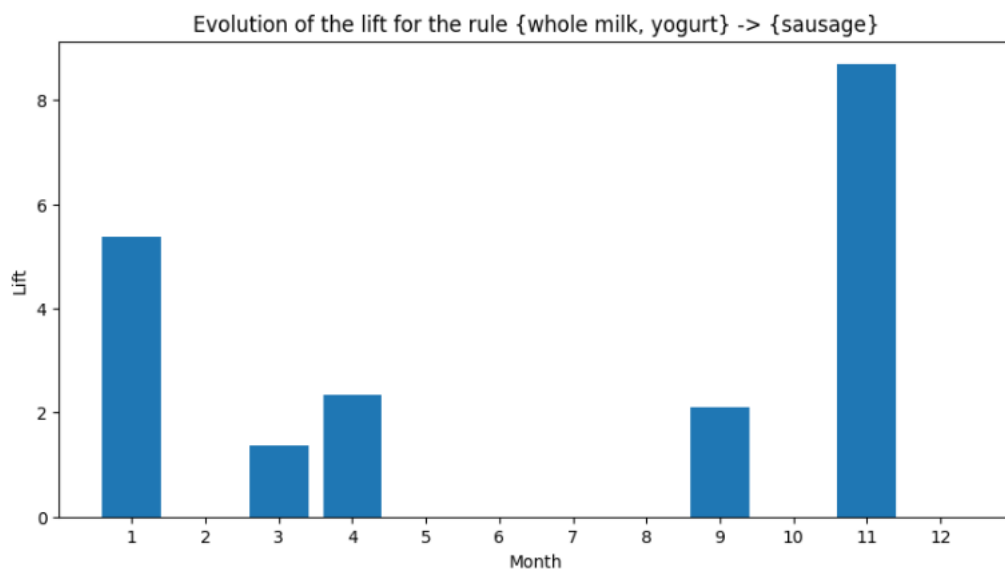


Figure 3.1. Monthly Evolution of the Lift Value for the Rule {whole milk, yogurt} -> {sausage}

Chapter 4. Discussion

4.1 Orange versus Python

This study used both Orange Data Mining and Python to perform Market Basket Analysis. The use of these two tools provided an opportunity to compare their respective strengths and limitations when applied to the same dataset.

Orange Data Mining was initially selected because of its visual interface and its accessibility for exploratory data analysis. The software facilitated the first stages of the project by allowing the dataset to be imported, prepared, and explored through a graphical workflow. After converting the product variables into categorical attributes, the Frequent Itemsets widget successfully generated frequent item sets, confirming the presence of recurring purchasing patterns within the dataset.

However, despite testing several support and confidence thresholds, the Association Rules widget did not generate any usable association rules. As discussed in Chapter 3, this limitation was mainly related to the sparse structure of the dataset and the large number of products. In addition, the minimum confidence threshold available in Orange may still have been too restrictive for the selected data. Although Orange provided a useful introduction to Market Basket Analysis and helped verify the suitability of the dataset, it was not sufficient for completing the objectives of this study.

Python offered greater flexibility and control over the analytical process. Using the pandas and mlxtend libraries, it was possible to perform the required data transformations, generate frequent item sets, extract association rules, and calculate support, confidence, and lift values. Unlike Orange, Python successfully produced association rules that could be interpreted and analysed.

An additional advantage of Python was the possibility of conducting a monthly analysis of association rules. By applying the Apriori algorithm separately to each month, it became possible to examine how strength of a selected rule evolved throughout the year. This analysis provided additional insights into the temporal variability of purchasing patterns that could not be obtained from the global analysis alone.

From a practical perspective, the comparison between the two tools highlights a trade-off between accessibility and analytical flexibility. In this study, this trade-off became particularly apparent during the rule generation stage. Orange provided an intuitive environment for data exploration and frequent itemset extraction, but it did not generate association rules despite several parameter adjustments. Python required more technical skills, yet its greater flexibility allowed the successful extraction of association rules and the implementation of the monthly analysis presented in Chapter 3.

Overall, the two tools played complementary roles in the research process. Orange was valuable for data exploration and initial validation of the dataset, whereas Python proved more effective for advanced analysis and result generation. The combination of both tools contributed to a better understanding of customer purchasing behaviour and allowed the research objectives of the study to be achieved.

4.2 Comparison with Literature

The findings of this study are broadly consistent with the existing literature on Market Basket Analysis and association rule mining. However, they also reveal several specific characteristics that provide additional insights into customer purchasing behaviour.

The methodological approach adopted in this study follows the framework proposed by Agrawal et al. (1993) and further developed by Agrawal and Srikant (1994). As in these studies, support, confidence, and lift were used to evaluate the strength of product associations. The results confirmed the usefulness of combining these indicators, as some association rules displayed relatively low support values but comparatively high lift values. This finding illustrates the importance of considering multiple evaluation measures rather than relying on support alone.

The frequent item sets identified in the dataset also support previous observations reported in retail-oriented studies. Products such as whole milk, yogurt, rolls/buns, and vegetables appeared among the most frequently purchased items, reflecting recurring purchasing patterns within grocery transactions. Martinez et al. (2021) similarly highlighted the

ability of Market Basket Analysis to identify frequently purchased products and reveal meaningful relationships between items.

The association rules generated in this study further illustrate the practical relevance of Market Basket Analysis discussed by Raorane et al. (2012). The identification of relationships between products demonstrates how association rule mining can help retailers better understand customer purchasing behaviour. Likewise, Hahsler et al. (2011) emphasized that association rules can support decision-making by revealing hidden patterns in transactional data. The repeated appearance of whole milk, yogurt, and sausage among the strongest associations identified in this study reflects this capability.

However, some findings also highlight characteristics that are less frequently emphasized in the literature. Although the rule {whole milk, yogurt} $\rightarrow$ {sausage} achieved one of the highest lift values in the dataset, its support remained relatively low. This result suggests that statistically strong associations do not necessarily correspond to frequently occurring purchasing behaviours. Such findings reinforce the importance of interpreting support, confidence, and lift jointly rather than independently.

The temporal analysis provides additional points of comparison with previous studies. Papavasileiou and Tsadiras (2011) argued that purchasing patterns may evolve over time and that temporal analyses can reveal relationships that remain hidden in global datasets. The results obtained in this study support this observation. While the selected rule appeared among the strongest associations in the global analysis, it was generated in only five months of the year and displayed substantial variations in lift values across periods.

Similarly, Kaur and Kang (2016) emphasized that customer purchasing behaviour is not necessarily stable over time. The strong differences observed between January, November, and the remaining months are consistent with this perspective. Rana and Mondal (2021) also noted that certain associations may emerge only during specific periods. The absence of the selected rule during seven months of the year provides additional evidence supporting this argument.

One possible explanation is that the purchasing relationship between these products may be influenced by seasonal consumption habits. However, the dataset does not provide information on promotions, holidays, or external events, making it impossible to confirm this interpretation. Consequently, any explanation of these temporal variations should be considered with caution.

Finally, the findings support the observations of Prasetio et al. (2025) , who emphasized the growing role of data-driven approaches in understanding customer behaviour and supporting decision-making. Although the present study relied on a single retail dataset, the identified associations and temporal variations demonstrate how transactional data can be transformed into actionable insights for retail analysis.

Overall, the results both confirm and extend previous findings reported in the literature. While the study supports the effectiveness of Market Basket Analysis for identifying product relationships, it also highlights the importance of considering temporal variations and interpreting association measures carefully, particularly when strong lift values are associated with relatively low support. These findings not only contribute to the academic understanding of Market Basket Analysis but also have practical implications for retail decision-making, which are discussed in the following section.

4.3 Managerial Implications

The results of this study provide several practical implications for retailers and managers seeking to improve decision-making through transactional data analysis.

First, the identification of frequent products such as whole milk, yogurt, rolls/buns, vegetables, and sausage may help retailers better understand customer purchasing habits. Products that are frequently purchased can receive particular attention in terms of inventory management, shelf organization, and stock availability to reduce the risk of shortages. Martinez et al. (2021) similarly emphasized that Market Basket Analysis can support retail decision-making by identifying recurrent purchasing patterns.

Second, the association rules identified in this study may support cross-selling strategies. The repeated relationships observed between whole milk, yogurt, and sausage suggest that these products may be purchased together more often than expected. Retailers could therefore consider placing related products in nearby locations or developing promotional campaigns involving complementary products. Raorane et al. (2012) highlighted that association rules can support product recommendations and cross-selling activities.

The results also highlight the importance of using multiple association measures simultaneously. The selected rule displayed a relatively low support value but a comparatively high lift value. This finding suggests that managers should not focus exclusively on frequently occurring combinations, as less common associations may still provide useful commercial information.

The temporal analysis provides additional managerial insights. The selected association rule was observed in only five months of the year, and its strength varied considerably across periods. These findings suggest that purchasing relationships may change over time and that retailers could benefit from monitoring association rules periodically rather than relying exclusively on a single global analysis. Similar observations were reported by Rana and Mondal (2021), who argued that certain associations may emerge only during specific periods.

More generally, the results illustrate the value of data-driven decision-making in retail environments. Transactional data can provide useful information about customer purchasing behaviour and support decisions related to product placement, promotions, inventory management, and sales strategies. Prasetio et al (2025) emphasized the growing importance of using analytical methods to support managerial decision-making, while Hahsler et al (2011) highlighted the practical usefulness of association rules for extracting actionable knowledge from transactional data.

Overall, the findings suggest that Market Basket Analysis can support retail decision-making by identifying purchasing patterns and revealing relationships between products. When combined with temporal analysis, these methods may help managers better understand changes in customer behaviour and adapt their strategies accordingly. The limitations of the present study and their implications for the interpretation of the results are discussed in the following section.

4.4 Limitations

Despite the contributions of this study, several limitations should be acknowledged when interpreting the results.

The first limitation concerns the dataset itself. The original source and the exact data collection context are not fully documented. The dataset was obtained from the Kaggle platform, and information regarding the retailer, geographical location, and data collection procedures is not available. In addition, the dataset does not contain customer characteristics such as age, gender, income, or household composition.

The available information is limited to transactional data and product purchases. The dataset does not provide information about product prices, promotions, holidays, marketing campaigns, or external events. Consequently, the factors influencing the observed purchasing patterns cannot be investigated directly, making the interpretation of temporal variations more difficult.

Another limitation concerns the analytical approach adopted in this study. The association rules were generated using the Apriori algorithm implemented in Python. Although Apriori is one of the most widely used algorithms in Market Basket Analysis (Agrawal et al. (1993; Agrawal and Srikant (1994))), other algorithms such as FP-Growth may produce slightly different results. Therefore, the findings may depend to some extent on the selected analytical method.

The association rules identified in the study also present certain limitations. Several rules displayed relatively low support values despite comparatively high lift values. Consequently, some statistically strong associations were observed in only a limited number of transactions. These findings should therefore be interpreted carefully when considering their practical significance.

The temporal analysis also presents important limitations. The monthly analysis focused on a single association rule, namely {whole milk, yogurt} $\rightarrow$ {sausage}. Although this rule was selected because it appeared among the strongest associations in the global analysis, other potentially relevant rules were not investigated. Consequently, the conclusions regarding temporal variations remain limited to this specific and cannot be generalized to all purchasing relationships within the dataset.

Finally, the study relied on a single retail dataset. Applying the same methodology to additional datasets from different retail environments could improve the external validity and generalizability of the findings.

Overall, these limitations do not invalidate the results of the study, but they should be considered when interpreting the findings and assessing their applicability to other retail contexts. They also provide several opportunities for future research, which are discussed in the following section.

4.5 Future Research

Several directions for future research can be proposed based on the limitations and findings of this study.

First, future studies could apply the same methodology to additional retail datasets obtained from different stores, regions, or market environments. Comparing results across multiple datasets would improve the generalizability of the findings and provide a broader understanding of customer purchasing behaviour.

Second, future research could investigate a larger number of association rules. The present study focused on the rule {whole milk, yogurt} $\rightarrow$ {sausage} for the temporal analysis because it appeared among the strongest associations in the global analysis. Examining additional rules could provide a more comprehensive understanding of the temporal evolution of purchasing of purchasing patterns.

Another possible extension would be the comparison of different association rule mining algorithms. Although this study employed the Apriori algorithm, future research could compare its performance with other methods such as FP-Growth to evaluate potential differences in the generated rules.

Future studies could also incorporate additional information related to customers and products. Variables such as age, gender, income, product prices, promotional campaigns, holidays, or external events could help explain purchasing behaviours and provide a deeper interpretation of temporal variations.

The integration of customer segmentation techniques may also represent an interesting research direction. Analysing purchasing patterns for different customer groups could reveal specific associations that are not observable in the overall population.

Beyond the limitations identified in this study, future research could also investigate the combination of Market Basket Analysis with predictive or recommendation approaches. Integrating association rules with other analytical techniques may provide additional insights into customer purchasing behaviour and support more advanced retail decision-making.

Overall, the present study demonstrates the potential of Market Basket Analysis while also identifying several opportunities for further investigation. Expanding the data sources, analytical methods, and variables considered may contribute to a more comprehensive understanding of purchasing behaviour in retail environments and support future developments in retail analytics.

General Conclusion

Summary of the Study

The objective of this study was to investigate customer purchasing behaviour using Market Basket Analysis techniques. More specifically, the research aimed to identify product associations from retail transactional data, compare the usefulness of Orange Data Mining and Python, analyse temporal variations in purchasing relationships, and examine the managerial implications of the obtained results.

To achieve these objectives, a retail transactional dataset obtained from Kaggle was analysed using both Orange Data Mining and Python. Orange was initially employed for exploratory analysis and frequent itemset generation, while Python was used to generate association rules and conduct the temporal analysis.

The study relied on the Apriori algorithm and the measures of support, confidence, and lift to evaluate relationships between products. In addition, a monthly analysis was performed to examine how the strength of a selected association rule evolved throughout the year.

Answers to the Research Questions

The first research question aimed to identify the product associations that could be extracted from retail transactional data through Market Basket Analysis. The results revealed several frequent products, including whole milk, yogurt, rolls/buns, vegetables, and sausage. Furthermore, the rule {whole milk, yogurt} $\rightarrow$ {sausage} emerged as one of the strongest associations identified in the global analysis.

The second research question examined the respective strengths and limitations of Orange Data Mining and Python. The results demonstrated that Orange was particularly useful for exploratory analysis and frequent itemset generation, whereas Python provided greater flexibility for association rule extraction and temporal analysis. The complementary use of both tools contributed to the successful completion of the study.

The third research question investigated how purchasing relationships evolved over time. The monthly analysis showed that the selected association rule was not stable throughout the year. The rule appeared in only five months and displayed considerable variations in lift values, with the strongest associations observed in January and November. These findings are consistent with the observations of Papavasileiou and Tsadiras (2011), Kaur and Kang (2016), and Rana and Mondal (2021), who emphasized the importance of considering temporal variations in purchasing behaviour.

The fourth research question concerned the implications of the findings for retail decision-making. As discussed in Section 4.3, the results suggest that Market Basket Analysis can support product placement, cross-selling strategies, inventory management, and the monitoring of purchasing relationships over time.

Contributions of the Study

This study provides both methodological and practical contributions.

From a methodological perspective, the research demonstrated the complementary use of Orange Data Mining and Python for Market Basket Analysis. The study also illustrates how temporal analysis can be incorporated into traditional association rule mining to investigate changes in purchasing behaviour over time.

From an empirical perspective, the findings showed that strong associations may present relatively low support values while maintaining high lift values. The results therefore emphasize the importance of interpreting support, confidence, and lift together rather than independently, which is consistent with the recommendations of Hahsler et al. (2011).

Finally, the study contributes to the existing literature by confirming that purchasing relationships may vary over time, as suggested by Rana and Mondal (2021). The results also demonstrate the value of combining global and temporal analyses to obtain a more complete understanding of customer purchasing behaviour.

Final Remarks

Overall, this study demonstrates the usefulness of Market Basket Analysis for understanding customer purchasing behaviour and identifying relationships between products. The findings show that transactional data can provide valuable information for both retail analysis and managerial decision-making.

Although several limitations were identified, the study contributes to the understanding of purchasing behaviour and provides practical insights for retail environments. Future research may extend this work by applying the methodology to additional datasets, investigating other association rules, comparing alternative algorithms, and incorporating supplementary variables that could further explain purchasing behaviour.

By combining association rule mining, temporal analysis, and multiple analytical tools, this study highlights the potential of Market Basket Analysis as a valuable approach for analysing retail transaction data and supporting data-driven decision-making.

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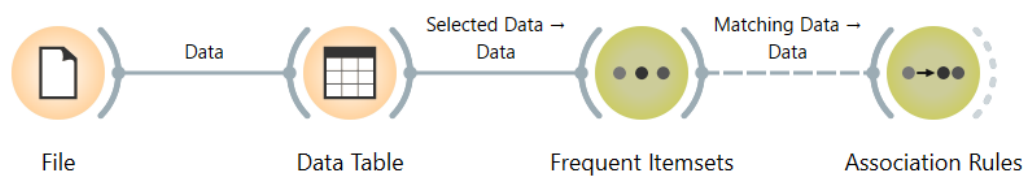
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Dataset reference:

<https://www.kaggle.com/code/chuarenyou/market-basket-analysis-mba/input>

Appendices



Appendix A. Orange Workflow

*** Frequent Itemsets - Orange

Info

Number of itemsets: 100000
 Selected itemsets: 0
 Selected examples: 0

Expand all Collapse all

Find itemsets

Minimal support: 0.001%
 Max. number of itemsets: 100000

Find Itemsets

Filter itemsets

Contains:

Min. items: 1 Max. items: 999

Apply these filters in search

Send Selection Automatically

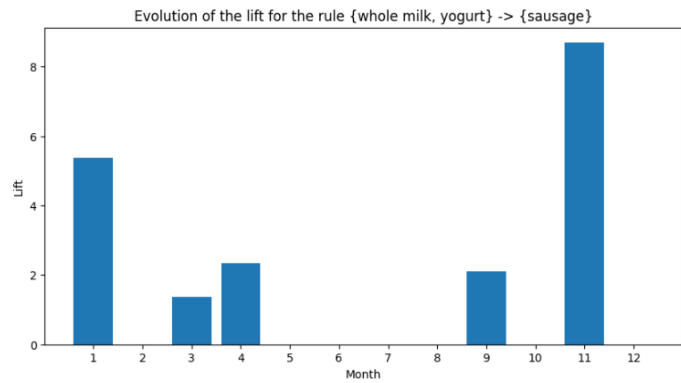
Itemsets	Support	%
Total général=2	10080	67.35
Total général=3	2698	18.03
whole milk=1	2232	14.91
Total général=2	1254	8.378
Total général=3	414	2.766
Total général=4	326	2.178
yogurt=1	153	1.022
Total général=2	40	0.2673
Total général=3	31	0.2071
Total général=4	35	0.2338
Total général=5	11	0.0735
Total général=6	6	0.04009
Total général=8	13	0.08686
Total général=7	11	0.0735
zwieback=1	1	0.006681
Total général=7	1	0.006681
Total général=5	92	0.6147
Total général=6	46	0.3073
Total général=8	43	0.2873
Total général=7	36	0.2405
zwieback=1	7	0.04677
Total général=3	2	0.01336
Total général=4	1	0.006681

Appendix B. Frequent Item Sets Generated in Orange

	antecedents	consequents	support	confidence	lift
1236	(sausage)	(whole milk, yogurt)	0.001470	0.024363	2.182917
1233	(whole milk, yogurt)	(sausage)	0.001470	0.131737	2.182917
1232	(whole milk, sausage)	(yogurt)	0.001470	0.164179	1.911760
1237	(yogurt)	(whole milk, sausage)	0.001470	0.017121	1.911760
474	(citrus fruit)	(specialty chocolate)	0.001403	0.026415	1.653762
475	(specialty chocolate)	(citrus fruit)	0.001403	0.087866	1.653762

Appendix C. Global Association Rules

	month	support	confidence	lift
0	1	0.004648	0.315789	5.364266
1	2	0.000000	0.000000	0.000000
2	3	0.001621	0.095238	1.366556
3	4	0.002451	0.136364	2.350832
4	5	0.000000	0.000000	0.000000
5	6	0.000000	0.000000	0.000000
6	7	0.000000	0.000000	0.000000
7	8	0.000000	0.000000	0.000000
8	9	0.001748	0.117647	2.102941
9	10	0.000000	0.000000	0.000000
10	11	0.003931	0.416667	8.688525
11	12	0.000000	0.000000	0.000000



Appendix D. Monthly Analysis Results

Appendix E. Python Code

```
import pandas as pd
from mlxtend.frequent_patterns import apriori, association_rules

df = pd.read_csv("Dataset 1 CSV.csv")

basket = (
    df.groupby(['Transaction_ID', 'itemDescription'])['itemDescription']
      .count()
      .unstack()
      .fillna(0)
)

basket = basket.map(lambda x: 1 if x > 0 else 0)
basket = basket.astype(bool)
```

E.1 Data Preparation

```
frequent_itemsets = apriori(
    basket,
    min_support=0.001,
    use_colnames=True
)

frequent_itemsets['length'] = (
    frequent_itemsets['itemsets']
    .apply(lambda x: len(x))
)
```

E.2 Global Apriori Analysis

```
rules = association_rules(
    frequent_itemsets,
    metric="confidence",
    min_threshold=0.001
)

table_42 = (
    rules.sort_values(
        by="lift",
        ascending=False
    )["antecedents",
      "consequents",
      "support",
      "confidence",
      "lift"]
    .head(6)
)
```

E.3 Association Rules

```
month1 = df["month"] == 1

basket_month1 = (
    month1.groupby(
        ['Transaction_ID',
         'itemDescription']
    )['itemDescription']
    .count()
    .unstack()
    .fillna(0)
)

basket_month1 = basket_month1.map(
    lambda x: 1 if x > 0 else 0
)

basket_month1 = basket_month1.astype(bool)
```

E.4 Monthly Analysis

```

results = []

for m in range(1, 13):
    data_month = df[df["month"] == m]

    basket_month = (
data_month.groupby(["Transaction_ID", "itemDescription"])["itemDescription"]
        .count()
        .unstack()
        .fillna(0)
    )

    basket_month = basket_month.map(lambda x: 1 if x > 0 else 0)
    basket_month = basket_month.astype(bool)

    frequent_itemsets_month = apriori(
        basket_month,
        min_support=0.001,
        use_colnames=True
    )

    rules_month = association_rules(
        frequent_itemsets_month,
        metric="confidence",
        min_threshold=0.001
    )

    selected_rule = rules_month[
(rules_month["antecedents"] == frozenset(["whole milk", "yogurt"])) &
(rules_month["consequents"] == frozenset(["sausage"]))
    ]

    if len(selected_rule) > 0:
        results.append({
            "month": m,
            "support": selected_rule.iloc[0]["support"],
            "confidence": selected_rule.iloc[0]["confidence"],

```

```
        "lift": selected_rule.iloc[0]["lift"]
    })
    else:
        results.append({
            "month": m,
            "support": 0,
            "confidence": 0,
            "lift": 0
        })

results_df = pd.DataFrame(results)
```

E.5 Monthly Association Rule Analysis